

NUMERICAL ANALYSIS

PROBLEM SET 4

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1. PROBLEM 1

1.a. We are deriving the 2-point open Newton-Coats formula

$$\int_a^b f(x)dx \approx I_{2,\text{open}}(f) = \frac{b-a}{3} \left(2f(a+h) - f(a+2h) + 2f(a+3h) \right)$$

for evenly spaced points with $h = \frac{b-a}{n+2}$. Now we set $n = 2$ so that $h = \frac{b-a}{4}$, and we need to sort out

$$I_{2,\text{open}}(f) = \int_a^b \sum_{i=0}^2 L_{2,i}(x) f(x_i) dx.$$

But we need to shift to an easier or simpler variable to work with. So we set $x = a + t \cdot h$, with $dx = h \cdot dt$. Now as x moves from a to b we see that ($b = a + 4h$), t moves from 0 to 4, and we have

$$\int_a^b f(x)dx = h \int_0^4 f(a+ht)dt,$$

which we interpolate $f(a+ht)$ at $t = 1, 2, 3$. We write out the Lagrangian polynomials $L_{2,0}, L_{2,1}, L_{2,2}$ for the sake of the problem. This follows as

$$L_{2,0}(x) = \frac{(t-2)(t-3)}{(1-2)(1-3)} = \frac{(t-2)(t-3)}{2} = \frac{1}{2}(t^2 - 5t + 6)$$

$$L_{2,1}(x) = \frac{(t-1)(t-3)}{(2-1)(2-3)} = -(t-1)(t-3) = -t^2 + 4t - 3$$